



MTL HACKATHON · DEVICE TRACK

Chilly Chiplet

Chip layout design GUI

That allows real-time thermal eval

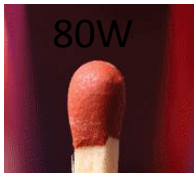
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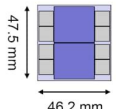
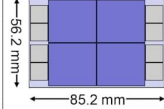
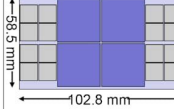
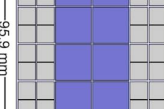
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Importance

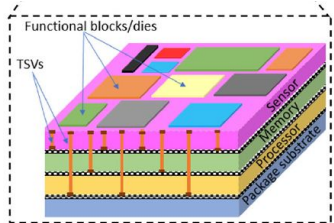
Heat on Chips is a huge issue!



Next-Generation GPU-HBM Roadmap : More GPU & HBM Integrated Above Interposer				
GPU Architecture	Rubin (2026)	Feynman (2029)	Post Feynman (2032)	Next-Gen Architecture (2035)
GPU Die Size	728 mm ²	750 mm ²	700 mm ²	600 mm ²
GPU Power	800 W	900 W	1,000 W	1,200 W
GPU-HBM Module	R200	F400	Post Feynman GPU-HBM Module	Next-Gen GPU-HBM Module
Interposer Size				
# of GPU Dies	×2	×4	×4	×8
# of HBM Stack	HBM4×8	HBM5×8	HBM6×16	HBM7×32
Interposer Die Size	2,194 mm ² (46.2 mm x 48.5 mm)	4,788 mm ² (85.2 mm x 56.2 mm)	6,014 mm ² (102.8 mm x 58.5 mm)	9,245 mm ² (96.4 mm x 95.9 mm)
Total Bandwidth	16 / 32 TB/s	48 TB/s	128/256 TB/s	1,024 TB/s
Total HBM Capacity	288/384 GB	400/500 GB	1,536/1,920 GB	5,120/6,144 GB
Total Power	2,200 W	4,400 W	5,920 W	15,360 W

HBM Roadmap Ver 1.7 Workshop by KAIST TERALAB

Compression to 2.5D/3D to increase yield/modularity exacerbate thermal issues (hidden warping/hotspot)!



3-D heterogeneous integration

Sharma & Ramos-Alvarado (2026)

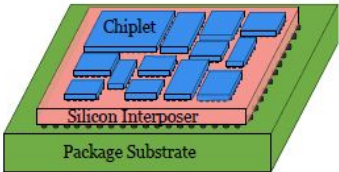
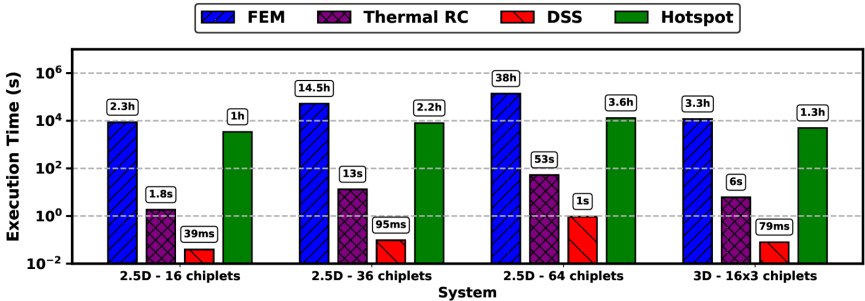


Fig. 2 Floorplan design of chiplets in 2.5D package.

Chen, et al (2023)

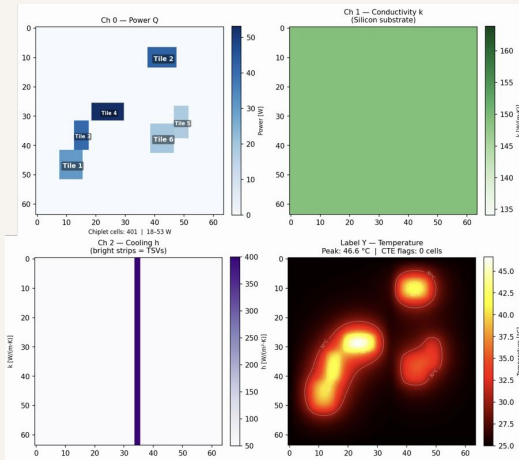
Time (2-30 hours each trial) for iterative FEM simulation to identify thermal issues!



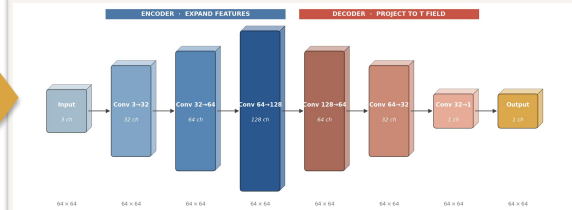
Pfromm, Kanani, Sharma, Solanki, Tervo, Park, Doppa, Pande, Ogras. MFIT: Multi-Fidelity Thermal Modeling for 2.5D and 3D Multi-Chiplet Architectures. arXiv:2410.09188 (2024).

Solution: builder GUI with AI surrogate trained on simulations

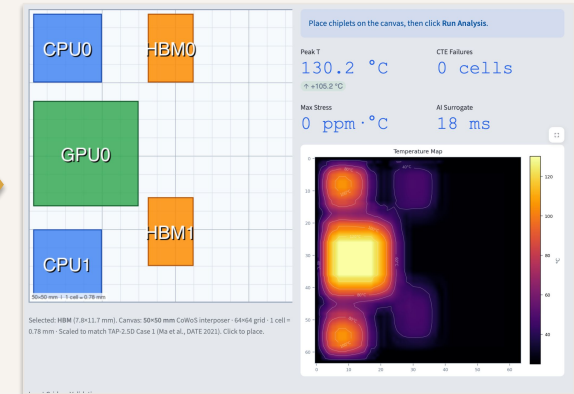
Physics Engine *data_generator.py*



ML surrogate *model_trainer.py*

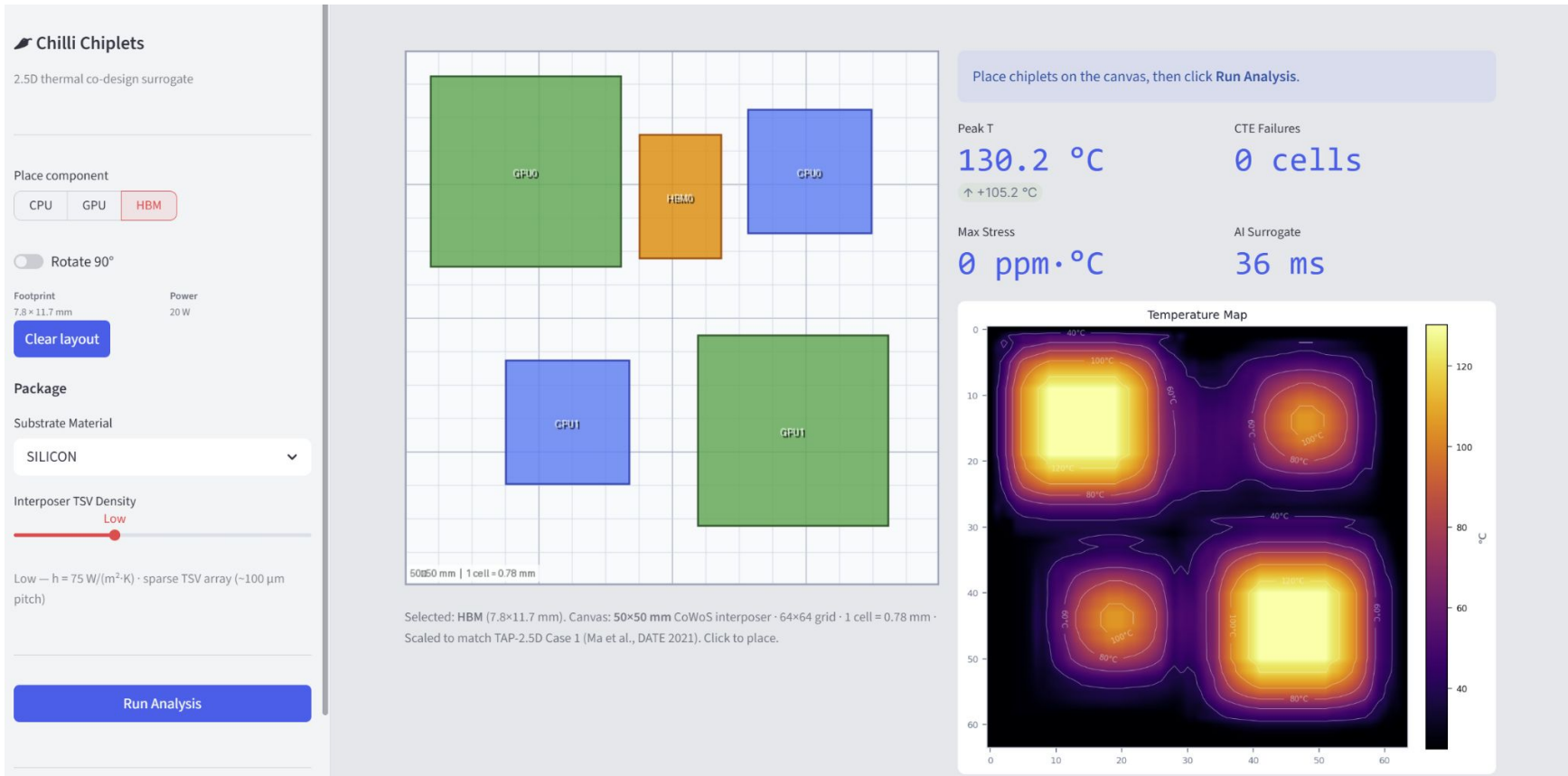


Engineer UI *app.py (Streamlit)*

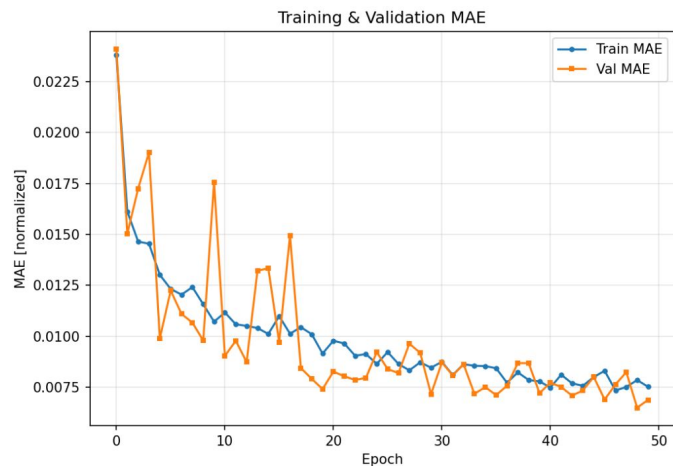


layout design+thermo+copilot query

Demo time



BENCHMARKING

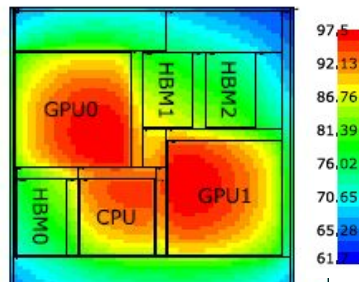
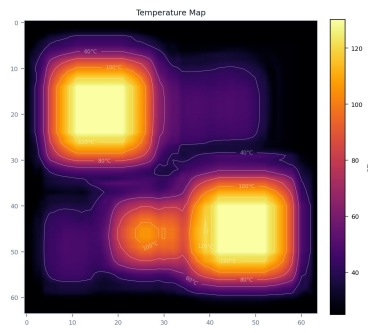


Our AI surrogate model
generalized our synthetic Data
(2.2 C av. error on 325 C span).
ML MODEL WORKS!!!

Heat map validation

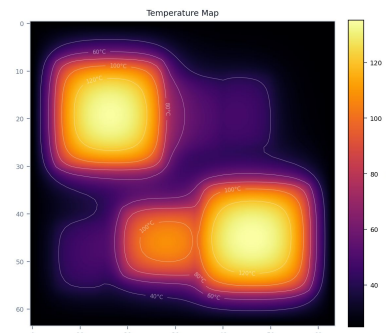
AI surrogate

Peak T = 130 C



Physics engine

Peak T = 135.3 C

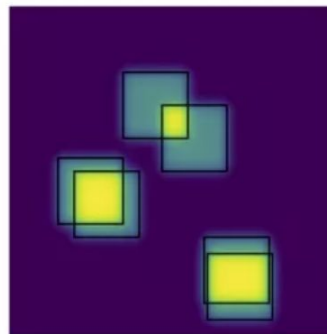


HotSpot simulation
(Ma, et al, 2021)
Peak T = 95.31 C

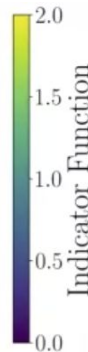
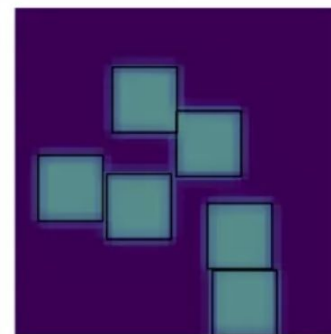
Next Step?



Before



After



Romano, et al (2025)

1. Replace our simple FDM with **HotSpot** as the data factory
 - a. production-grade fidelity, same pipeline
2. Add a layout optimizer algorithm
 - a. Optimize layout based on chiplet **thermal issues** and **communication latency** and → **conflicting hardware constraints**

References

- Romano, G., Jain, A., Dehmamy, N., Chi, C., & Zhang, X. *DiffChip: Thermally Aware Chip Placement with Automatic Differentiation*. arXiv:2502.16633, 2025.
- Ma, Y., Delshadtehrani, L., Demirkiran, C., Abellán, J. L., & Joshi, A. *TAP-2.5D: A Thermally-Aware Chiplet Placement Methodology for 2.5D Systems*. DATE 2021.
- Graening, A., Gupta, P., Kahng, A. B., Pramanik, B., & Wang, Z. *ChipletPart: Cost-Aware Partitioning for 2.5D Systems*. arXiv:2507.19819, 2025.
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- Sharma, M. K., & Ramos-Alvarado, B. *Thermal Management of 3-D Heterogeneously Integrated Microelectronics: Challenges and Future Research Directions*. Communications Engineering (Nature Portfolio), 2026.
- Huang, W. et al. *HotSpot: A Compact Thermal Modeling Methodology for Early-Stage VLSI Design*. IEEE TVLSI 2006.